

# Ruilin (Peter) Xie

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## Education

### University of California, Irvine

Irvine, CA

*Master of Software Engineering*

Sep 2024 – Dec 2025

Coursework: Data Structures/Algorithms, Distributed Architecture, Software Testing & Debugging, Reverse Engineering, Concurrent/Network Programming

### University of California, Irvine

*Bachelor of Arts, Business Economics*

Jun 2018 – Aug 2022

## Skills

**Languages:** C# · C++ · Python · Blueprint (UE5) · Java · TypeScript

**Engines & Tools:** Unity 2022 · Unreal Engine 5 · UMG · Git · Diversion

**Disciplines:** Gameplay Systems · Runtime Architecture · Data-Driven Design · Technical Design · Balance Tooling

## Projects

### CrossBolt — Rhythm Game Prototype

Unity 2022, C#, | Team (4) | 2025

[github.com/MisakaRinOwO/Crossbolt-Code-Samples](https://github.com/MisakaRinOwO/Crossbolt-Code-Samples)

- Designed data-driven note schema, chart addressing, and custom .txt grammar consumed by runtime loader and editor tooling
- Implemented chart pipeline: NoteLevelConverter → typed timestamp queue → ChartSpawner with type-aware prefab selection and O(1) hit lookup
- Built Unity Editor beat-map editor; optimized the main judgement path from O(N) active-list scan to O(P) pending-window checks + O(1) hold lookups

### Factory Experiment — Data-Oriented Factory Simulation

UE5.6, C++, Blueprint | Solo | 2026

[github.com/MisakaRinOwO/UE5-Factory-Experiment](https://github.com/MisakaRinOwO/UE5-Factory-Experiment)

- Built sparse chunked world storage with dense 32×32 cell arrays, negative-coordinate support, rotated footprints, occupancy validation, and read/write access separation
- Implemented coord-keyed conveyor logistics and HISM presentation without Actor-per-segment simulation; snapshot/planning transfers prevent traversal-order conflicts
- Developed a fixed-step Miner→Conveyor→Smelter→Storage loop, with Data Asset-authored buildings, recipes, resources, runtime ports, and live UMG inspection

### Void Architect — Modular Ship Combat Prototype

UE5, Blueprint, Python | Solo | 2026

- Implemented tick-driven combat simulation: tracking vs. agility hit probability, weapon cooldowns, shield→HP overflow, and replay event recording
- Built equilateral-triangle placement grid with orientation-aware adjacency rules; data-driven module architecture via UE5 Data Assets with CPU/power budget constraints
- Python aggregation scripts identified Sniper shield outlier; nerfed values and verified win-rate spread across full build matrix with replay UI inspection

### Project Summon — Action RPG Capstone

UE5, Blueprint, C++ | Team (5) | 2025

- Shipped complete gameplay loop (hub → room combat → loot → return) on fixed 10-week schedule; owned room progression, combat, inventory, and persistence systems
- Built BP\_RoomManager as room authority: overlap-scoped enemy tracking, O(1) clear check, and enemy/chest/portal spawning on clear trigger
- Integrated teammate C++ enemy code by exposing HP/TakeDamage (UPROPERTY/UFUNCTION) and implemented GameInstance-based cross-level player data sync

### Godot Engine — Open Source C++ Contributions

C++, Godot 4 | 2025

[github.com/godotengine/godot/pulls?q=is%3Apr+author%3AMisakaRinOwO](https://github.com/godotengine/godot/pulls?q=is%3Apr+author%3AMisakaRinOwO)

- PR #107101 fixed CSGShapes3D hover feedback; PR #107296 added 3D Ruler helper lines/labels; both passed CI and maintainer review